BCS-087	Artificial Intelligence and Expert System	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To learn about the knowledge of intelligent agents
CO2	To learn about machine learning
CO3	Understand Bayesian network and fuzzy logic in case of uncertainty.
CO4	To understand HMM model
CO5	To understand Uniformed search strategies

Detailed Syllabus

MODULE-I

Introduction: Introduction to Artificial Intelligence, Foundations and History of Artificial Intelligence, Applications of Artificial Intelligence, Intelligent Agents, Structure of Intelligent Agents. Computer vision, Natural Language Possessing.

Introduction to Search: Searching for solutions, Uniformed search strategies, Informed search strategies, Local search algorithms and optimistic problems, Adversarial Search, Search for games, Alpha – Betapruning.

MODULE-II

Knowledge Representation & Reasoning: Propositional logic, Theory of first order logic, Inference in First order logic, Forward & Backward chaining, Resolution, Probabilistic reasoning, Utility theory, Hidden Markov Models (HMM), Bayesian Networks.

MODULE-III

Machine Learning: Supervised and unsupervised learning, Decision trees, Statistical learning models, learning with complete data - Naive Bayes models, Learning with hidden data –EM algorithm, Reinforcement learning, Expert systems architecture. Generalities about expert systems. Conceptual infrastructure of expert systems.

Course Outcomes: After the completion of the course the student will be able to:

CO1	To apply the knowledge of intelligent agents and the heuristic search techniques.
CO2	To analyze the role of knowledge representation techniques such as propositional and predicate logic in AI.
CO3	To apply the Bayesian network and fuzzy logic in case of uncertainty.
CO4	To analyze different types of planning and learning techniques.
CO5	To apply Uniformed search strategies

Text Books:

1. Stuart Russell, Peter Norvig, "Artificial Intelligence – A Modern Approach", PearsonEducation.
2. Elaine Rich and Kevin Knight, "Artificial Intelligence", McGraw-Hill

Reference Books:

1. E Charniak and D McDermott, "Introduction to Artificial Intelligence", PearsonEducation.
2. Dan W. Patterson, "Artificial Intelligence and Expert Systems", Prentice Hall of India.